Practical No. 8: Intrusion Detection System (IDS)

Aim

To **set up and configure an Intrusion Detection System (IDS)** to monitor network traffic and detect potential security breaches or suspicious activities.

Objectives

- To understand the concept of **Intrusion Detection System (IDS)**.
- To study different types of IDS.
- To understand how an IDS monitors network traffic.
- To configure a basic IDS using **Snort**.
- To create and apply a simple detection rule.
- To observe and analyse alerts generated by the IDS.

Theory

1. Intrusion Detection System

- An **Intrusion Detection System (IDS)** is a security mechanism that monitors network traffic or system activities to identify suspicious, malicious, or unauthorized activities.
- An IDS analyses the observed activity and generates an **alert** when the activity matches a known attack pattern or suspicious behaviour.

2. Purpose of IDS

- Monitor network traffic.
- Detecting suspicious activities.
- Identify possible attacks.
- Generate security alerts.
- Help administrators investigate security incidents.

3. Types of IDS

A. Network-Based IDS (NIDS)

- A **Network-Based Intrusion Detection System (NIDS)** monitors network traffic passing through a network.

Example:
- Snort, Suricata etc

B. Host-Based IDS (HIDS)

A **Host-Based Intrusion Detection System (HIDS)** monitors activities on an individual computer or server.

It can monitor:

- System logs
- File changes
- Login activities
- Processes
- System events

4. Detection Techniques

IDS commonly uses two major detection approaches.

A. Signature-Based Detection

- Detects known attacks using predefined signatures or rules.
- Effective for known attack patterns.
- Requires updated signatures to detect newly identified threats.

B. Anomaly-Based Detection

- Establishes a baseline of normal behaviour.
- Detects activities that significantly differ from normal behaviour.
- Can help identify previously unknown patterns.

5. Snort

Snort is a widely used network intrusion detection and prevention tool.

It can:

- Capture network traffic.
- Analyse packets.
- Apply detection rules.
- Generate alerts.
- Log suspicious network activity.

6. IDS Rule

- IDS rules define the conditions under which an alert should be generated.
- For example, a simple ICMP detection rule can be written as:
- alert icmp any any -> any any (msg:"ICMP Traffic Detected"; sid:1000001;)

Procedure: Configure IDS Using Snort

Procedure

Part A: Installing and Verifying Snort

Step 1: Install Snort

1. Download the appropriate Snort package for the Windows laboratory system.
2. Install Snort using the installation wizard.
3. Note the Snort installation directory.

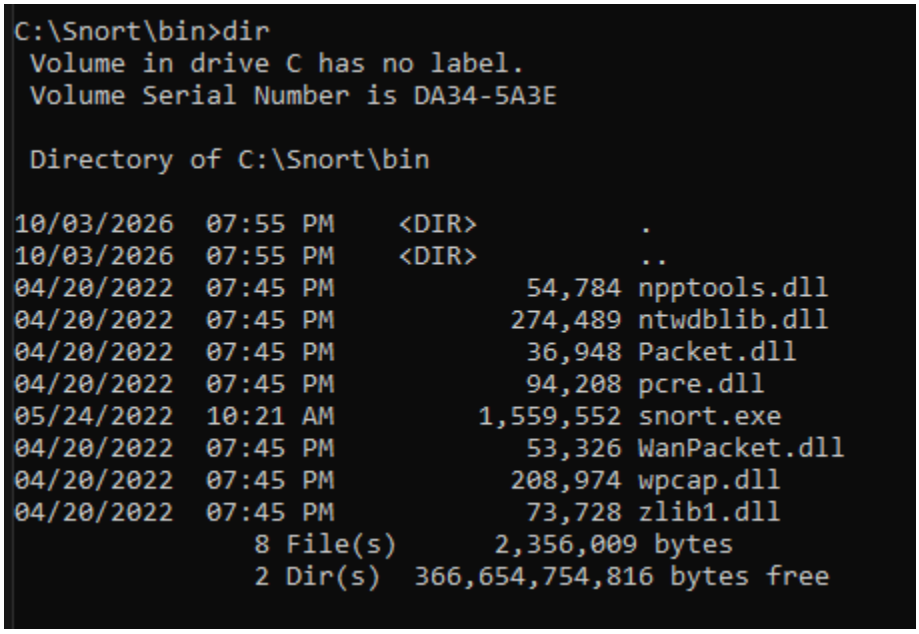
For example:

- C:\Snort
- The Snort executable may be located in:
- C:\Snort\bin

Step 2: Open Command Prompt

Open:

- **Start → Command Prompt → Run as Administrator**
- Navigate to the Snort directory:
- cd C:\Snort\bin



Step 3: Verify Snort Installation

Run:

- snort -V
- If Snort is installed correctly, the version information will be displayed.

Example: Snort Version 2.9.x

- If Windows displays:
- 'snort' is not recognized as an internal or external command
- navigate to the directory containing snort.exe and execute Snort from that directory.

Part B: Identifying the Network Interface

Step 4: Display Network Configuration

Run:

- ipconfig
- Identify the active network adapter.

For example:

```
C:\Snort\bin>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1090:fffc:e014:dd32%4
   IPv4 Address. . . . . . . . . . . : 192.168.0.164
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.0.1
```

The active adapter will be used for packet monitoring.

Step 5: Display Snort Network Interfaces

For Snort on Windows, run:

- snort -W
- Snort displays the available network interfaces.

Example:

Index Interface

1 \Device\NPF_{...}

2 \Device\NPF_{...}

Note the interface number corresponding to the active network connection.

Part C: Testing Packet Monitoring

Step 6: Start Packet Monitoring

Suppose the required interface number is 2.

Run:

- snort -v -i 2
- Replace 2 with the interface number identified on your computer.
- Snort will start displaying captured network traffic.

Step 7: Generate Test Traffic

Open another Command Prompt.

- In an authorized laboratory/test network, generate ICMP traffic using:
- ping <test-system-IP>
- For example:
- ping 192.168.1.20
- Observe the Snort Command Prompt.
- You should see packets being captured.
- Press:
- Ctrl + C
- to stop packet monitoring.

Part D: Creating an IDS Rule

Step 8: Create local.rules

- Go to the Snort rules directory.
- For example:
- C:\Snort\rules
- Create a file named:
- local.rules
- Open the file using Notepad and enter:

- alert icmp any any -> any any (msg:"ICMP Traffic Detected"; sid:1000001;)
- Save the file.

Part E: Configure Snort

Step 9: Open snort.conf

- Locate the Snort configuration file, for example:
- C:\Snort\etc\snort.conf
- Open it using a text editor.
- Ensure that the custom rule file is included in the configuration:
- include $RULE_PATH/local.rules
- Save the configuration file.